Rolling a die

Probability of getting 1

$$\frac{1}{6}$$



Rolling a die

Probability of getting 2

$$\frac{1}{6}$$



Rolling a die

Outcome	Probability
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Rolling a die

Outcome	Probability
1	0.17
2	0.17
3	0.17
4	0.17
5	0.17
6	0.17
All else	0



3 What is a distribution_

Rolling a die

Outcome	Probability
1	0.17
2	0.17
3	0.17
4	0.17
5	0.17
6	0.17
All else	0



What is a distribution_

Visual representation

Uniform distribution



What is a distribution_

Rolling two dice

Possibilities?

(1,1) (2,1) (3,1) (4,1) (5,1) (6,1)
(1,2) (2,2) (3,2) (4,2) (5,2) (6,2)
(1,3) (2,3) (3,3) (4,3) (5,3) (6,3)
(1,4) (2,4) (3,4) (4,4) (5,4) (6,4)
(1,5) (2,5) (3,5) (4,5) (5,5) (6,5)
(1,6) (2,6) (3,6) (4,6) (5,6) (6,6)



Rolling two dice

Probability of getting 1?

0



Rolling two dice

Probability of getting 2?

$\frac{1}{36}$ or 0.03



Rolling two dice

Probability of getting 3?

(1,2) and (2,1)

$$\frac{2}{36} \text{ or } 0.06$$



Number of favorable outcomes divided by the total number of outcomes

8 What is a distribution_

Rolling two dice

Outcome	Probability
2	0.03
3	0.06
4	0.08
5	0.11
6	0.14
7	0.17
8	0.14
9	0.11
10	0.08
11	0.06
12	0.03
	0

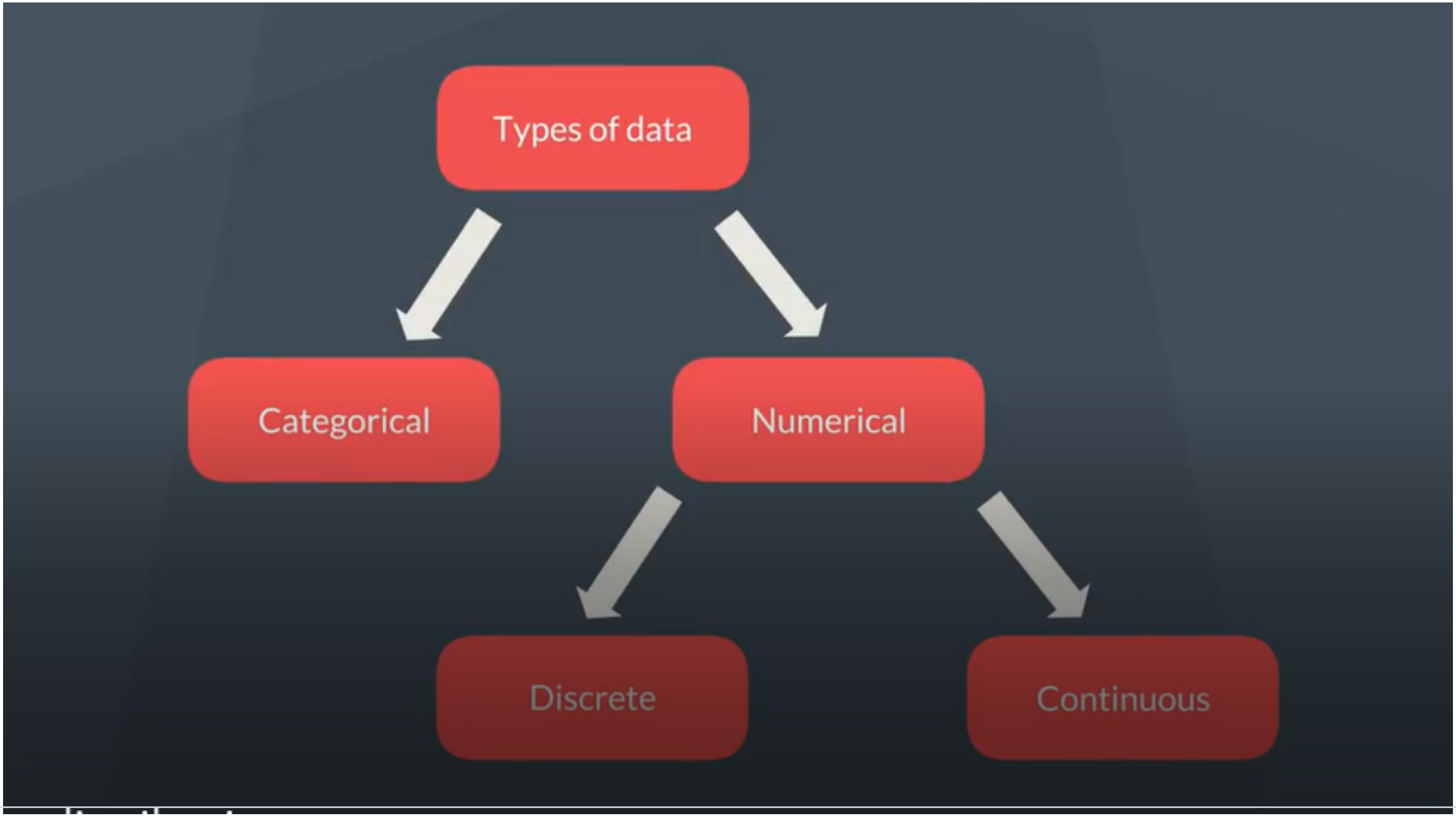


What is a distribution_

Visual representation



istribution_


Types of data
Categorical
Numerical
Discrete
Continuous